UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Parallel and multiprocessor architecture

a) How classification of parallel processor is constructed?
b) Give ONE (1) example for each parallel processor architecture.

Source: CSC429 (8).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Computer generations and history

Question 1 is based on the following placard in the Computer History Museum.
The Jacquard System was developed by
Joseph Marie Jacquard to automatically
produce the intricate woven patterns.
The deck of punched cards that
controls the Machine X is located at
the left center on the far side of the
figure. In the early days of software
piracy, the intricate fabric design card
decks would be stolen by competing
textile mills.
Figure 1: Machine X
a) Name the Machine X in Figure 1.
b) In which computer generation was Machine X developed?
c) Name the possible computer component that has similar function as the deck of punch card in Machine X.
d) Identify the input and output for Machine X.

Source: CSC429 (6).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Data representation and number systems

Convert 3.25 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

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Answer: 3.2510 = 0x40500000 (sign=0, exponent=10000000, fraction=10100000000000000000000)

- *1 mark for the sign bit.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the correct 32-bit grouping.*
- *1 mark for the final hexadecimal value.*

QUESTION A5 (10 marks)

Topic: Input/output and interrupts

a) Explain the problem in polling 1/0 method.
b) Explain how buffering can help to improve the device overall performance?
c) Discuss how DMA technique can help the 1/0 transfer performance instead of using polling method and interrupt 1/0 method?

Source: CSC429 (10).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Data representation and number systems

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: 63 - (-41)

a) Calculate 63 - (-41) using sign and magnitude.

Answer: left 63 -> 0 0111111; right -41 -> 1 0101001; 63 - (-41) = 104 -> sign-magnitude 0 1101000

- *1 mark for operand representations.*
- *1 mark for sign handling.*
- *1 mark for magnitude arithmetic.*
- *1 mark for the result sign bit.*
- *1 mark for the final byte.*

b) Calculate 63 - (-41) using 1's and 2's complement.

Answer: left 63 -> 1's 00111111, 2's 00111111; right -41 -> 1's 11010110, 2's 11010111; 63 - (-41) = 104 -> 1's 01101000, 2's 01101000

- *1 mark for the positive operand in binary.*
- *1 mark for the 1's complement of the negative operand.*
- *1 mark for adding 1 to form the 2's complement.*
- *1 mark for the correct 8-bit result.*
- *1 mark for converting the result back to decimal.*

QUESTION A7 (10 marks)

Topic: Computer architecture and CPU organisation

a) Discuss the function of each Central Processing Unit (CPU) component'
b) Elaborate on what will happen in the machine cycle if the bus system is faulty

Source: CSC429 (20).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Input/output and interrupts

Question 5 is based on the following scenario.
"Nurathia Farhana browses the UiTM4fly.com website to book for a return flight from KLIA to LAX. While comparing for a better price, two messages popped up in her screen. Figure 1 (a) and (b) illustrate the conveyed messages.
A hardware failure has been
found. Please restart your
computer now.
Do you want to be upgraded
to business class now?

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(b)
(a)
Figure 1: The pop-up messages
a) Which message is the non-maskable interrupt? Justify your answer by giving appropriate reason.
b) Differentiate between maskable and non-maskable interrupt?
c) Outline the common way to handle multiple interrupts as happened in the given scenario.

Source: CSC429 (11).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Computer architecture and CPU organisation

a) Compare system interconnection and CPU interconnection in terms of:
i) function
ii) physical appearance.
b) Describe the function of the THREE (3) different bus types in a computer system.

Source: CSC429 (16).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Memory hierarchy and cache

a)
Discuss the main function of cache memory.
b)
How can the data stored in the cache and main memory be synchronized.
c)
Contrast between SRAM and DRAM based on the following characteristics:
Type of storage
Availability of charges leak
price
Example of location of memory usage

Source: CSC429 (15).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 5 and Table 1
z-A+c2/B*D-3A
Figure 5
Write an appropriate program using conect assembly language notations to compute the equation shown in Figure 5 using instructions for zero-address, one-address, and three-address. The available instructions are provided in Table 1. The registers labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register z.
Table 1: lt/achine instruction
Three-address
Zero-address
One-address
ADD (p,q,r)
LOAD p
PUSH p
SUB (p,q,r)
STOR p
POP p
MUL (p,q,r)
ADD p
ADD
DIV (p,q,r)
SUB p
SUB
MUL p
MUL
DIV p
DIV

Source: CSC429 (21).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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